

Theory of Automata (BCS-402)

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Unit – IV

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Push Down Automata (PDA)

- Regular languages are those that are accepted by finite automata.
- But there is a class of automata that can be associated with context free languages.
- As we have seen, finite automata cannot recognize all context free languages because finite automata have strictly finite memories, whereas the recognition of context free language may require storing an unbounded amount of information.
- This suggest that we might try a stack as a storage mechanism, allowing unbounded storage that is restricted to operating like a stack. This gives us a class of machines called **push down automata (PDA)**.

Push Down Automata (PDA)

- Let us consider the components of the push down automata and the way it operates.

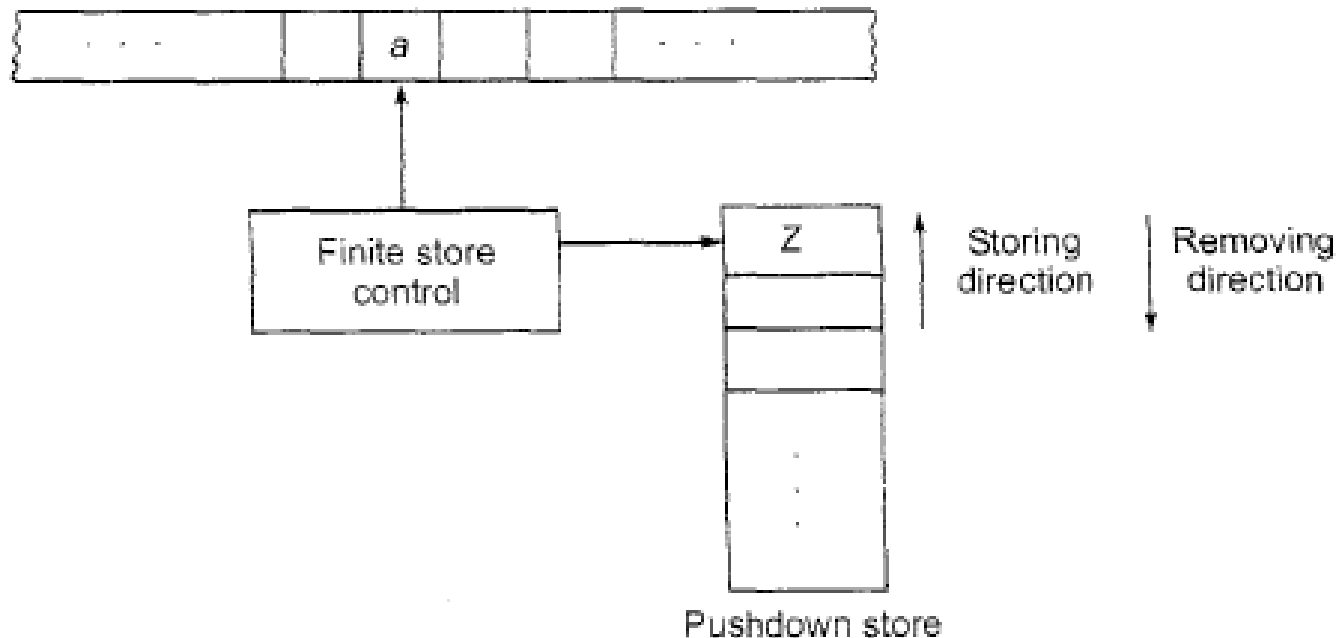


Fig. 7.1 Model of a pushdown automaton.

Push Down Automata (PDA)

- It has a read only input tape, input alphabet, finite state control, a set of final states, and an initial state as in the case of a finite automata.
- In addition to these, it has stack called the push down store (PDS).
- It is a read-write push down store as we add elements to PDS or remove elements from PDS.
- A finite automaton is in some state and on reading an input symbol moves to a new state.
 - The PDA is also in some state and on reading an input symbol and a topmost symbol in PDS moves to a new state and writes a string of symbols in PDS.

Push Down Automata (PDA)

- **Definition:**

- A **Push Down Automata** is defined by seven tuple $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ where

Q = finite set of internal states

Σ = input alphabets

Γ = finite set of push down symbols

δ = the transition function from $Q \times (\Sigma \cup \{\Lambda\}) \times \Gamma$ to the set of finite subsets of $Q \times \Gamma^*$ i.e., $Q \times (\Sigma \cup \{\Lambda\}) \times \Gamma \rightarrow Q \times \Gamma^*$

$q_0 = q_0 \in Q$ is the initial state of the control unit

Z_0 = special push-down symbol called the initial symbol on the push-down store

$F = F \subseteq Q$ is the set of final states.

Push Down Automata (PDA)

- The argument of δ are current state of the control unit, the current input symbol and current symbol on top of the stack.
- The result is a set of pairs (q, x) where q is the next state and 'x' is a string which is put on top of the stack in place of the single symbol there before.
- Note that the second argument of δ maybe Λ indicating that the move does not consume an input symbol is possible. We will call such move a Λ -transition.

For example, suppose the set of transition rules in PDA contains

$$\delta(q_1, a, b) = (q_2, a) \quad \text{-----(i)}$$

$$\delta(q_1, b, a) = (q_1, \Lambda) \quad \text{-----(ii)}$$

Here in exp. (ii) PDA goes into state q_1 and symbol 'a' is removed from the stack.

In exp. (i) PDA goes into state q_2 and symbol 'a' replace 'b' on top of stack.

Applications of PDA

- PDA's are the devices for accepting the context free languages.
- It can also store unbounded amount of information.
- PDA's are used in parsing for identifying valid sequence of words in compiler.

Instantaneous Description (ID) of PDA

Definition 7.2 Let $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a pda. An instantaneous description (ID) is (q, x, α) , where $q \in Q$, $x \in \Sigma^*$ and $\alpha \in \Gamma^*$.

For example, $(q, a_1a_2 \dots a_n, Z_1Z_2 \dots Z_m)$ is an ID. This describes the pda when the current state is q , the input string to be processed is $a_1a_2 \dots a_n$. The pda will process $a_1a_2 \dots a_n$ in that order. The PDS has $Z_1, Z_2, \dots, Z_m$ with Z_1 at the top, Z_2 is the second element from the top, etc. and Z_m is the lowest element in PDS.

Initial ID:

Definition 7.3 An initial ID is (q_0, x, Z_0) . This means that initially the pda is in the initial state q_0 , the input string to be processed is x , and the PDS has only one symbol, namely Z_0 .

Note: In an ID (q, x, α) , x may be Λ . In this case the pda makes a Λ -move.

Move Relation

Definition 7.4 Let A be a pda. A move relation, denoted by $\vdash$, between IDs is defined as

$$(q, a_1 a_2 \dots a_n, Z_1 Z_2 \dots Z_m) \vdash (q', a_2 a_3 \dots a_n, \beta Z_2 \dots Z_m)$$

if $\delta(q, a_1, Z_1)$ contains (q', β) .

If $\beta = Y_1 Y_2 \dots Y_k$, then Fig. 7.2 illustrates the move relation.

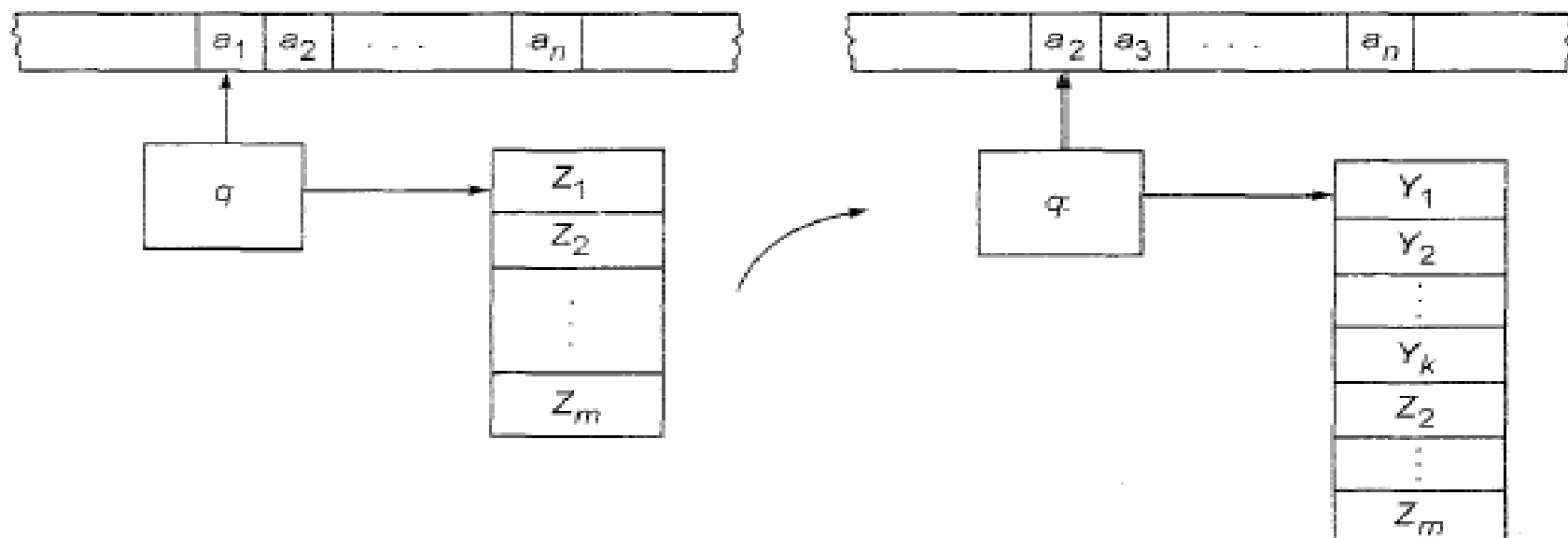


Fig. 7.2 An illustration of the move relation.

Move Relation

Remark As $\vdash$ defines a relation in the set of all IDs of a pda, we can define the reflexive-transitive closure $\vdash^*$ which represents a definite sequence of n moves, where n is any non-negative integer.

Note: When we deal with more than one pda, we also specify the pda while describing the move relation. For example, a move relation in A is denoted by $\vdash_A$.

Move Relation

Example:

$$A = (\{q_0, q_1, q_f\}, \{a, b, c\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \{q_f\})$$

is a pda, where δ is defined as

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}, \quad \delta(q_0, b, Z_0) = \{(q_0, bZ_0)\} \quad (7.5)$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}, \quad \delta(q_0, b, a) = \{(q_0, ba)\} \quad (7.6)$$

$$\delta(q_0, a, b) = \{(q_0, ab)\}, \quad \delta(q_0, b, b) = \{(q_0, bb)\} \quad (7.7)$$

$$\begin{aligned} \delta(q_0, c, a) = \{(q_1, a)\}, \quad \delta(q_0, c, b) = \{(q_1, b)\}, \quad \delta(q_0, c, Z_0) \\ = \{(q_1, Z_0)\} \end{aligned} \quad (7.8)$$

$$\delta(q_1, a, a) = \delta(q_1, b, b) = \{(q_1, \Lambda)\} \quad (7.9)$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_f, Z_0)\} \quad (7.10)$$

Move Relation

We can explain the concepts of ID, moves, etc. for this pda A . Suppose the input string is $acab$. We will see how the pda processes this string. An initial configuration is $(q_0, bacab, Z_0)$. We get the following moves:

$$\begin{array}{ll}
 (q_0, bacab, Z_0) \vdash (q_0, acab, bZ_0) & \text{by Rule (7.5)} \\
 \vdash (q_0, cab, abZ_0) & \text{by Rule (7.7)} \\
 \vdash (q_1, ab, abZ_0) & \text{by Rule (7.8)} \\
 \vdash (q_0, b, bZ_0) & \text{by Rule (7.9)} \\
 \vdash (q_1, \Lambda, Z_0) & \text{by Rule (7.10)} \\
 \vdash (q_f, \Lambda, Z_0) & \text{by Rule (7.10)}
 \end{array}$$

i.e.

$$(q_0, bacab, Z_0) \vdash^* (q_f, Z_0)$$

Proceeding in a similar way, we can show that

$$(q_0, wcw^T, Z_0) \vdash^* (q_f, \Lambda, Z_0) \text{ for all } w \in \{a, b\}^*$$

Move Relation

Suppose an initial configuration is $(q_0, abcb b, Z_0)$. Then, we have

$$(q_0, abcb b, Z_0) \vdash (q_0, bcb b, aZ_0) \quad \text{by Rule (7.5)}$$

$$\vdash (q_0, cb b, baZ_0) \quad \text{by Rule (7.6)}$$

$$\vdash (q_1, bb, baZ_0) \quad \text{by Rule (7.8)}$$

$$\vdash (q_1, b, aZ_0) \quad \text{by Rule (7.9)}$$

Once the pda is in ID (q_1, b, aZ_0) , it has to halt as $\delta(q_1, b, a) = \emptyset$. Hence, we have

$$(q_0, abcb b, Z_0) \vdash^* (q_1, b, aZ_0)$$

As $\delta(q_0, c, Z_0) = \emptyset$, the pda cannot make any transition if it starts with an ID of the form (q_0, cw, Z_0) .

Deterministic PDA

Definition 7.5 A pda $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ is deterministic if (i) $\delta(q, a, Z)$ is either empty or a singleton, and (ii) $\delta(q, \Lambda, Z) \neq \emptyset$ implies $\delta(q, a, Z) = \emptyset$ for each $a \in \Sigma$.

Consider the pda given in Example 7.2. $\delta(q, a, Z)$ given by Rules (7.5)–(7.10) are singletons. Also, $\delta(q_1, a, Z_0) = \emptyset$ and $\delta(q_1, a, Z_0) = \emptyset$ for all $a \in \Sigma$. So the pda given in Example 7.2 is deterministic.

Acceptance by PDA

A pda has final states like a nondeterministic finite automaton and has also the additional structure, namely PDS. So we can define acceptance of input strings by pda in terms of final states or in terms of PDS.

Acceptance by PDA by final state:

Definition 7.6 Let $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a pda. The set accepted by pda by final state is defined by

$$T(A) = \{w \in \Sigma^* | (q_0, w, Z_0) \xrightarrow{*} (q_f, \Lambda, \alpha) \text{ for some } q_f \in F \text{ and } \alpha \in \Gamma^*\}$$

Acceptance by PDA

Example:

Construct a pda A accepting $L = \{wcw^T \mid w \in \{a, b\}^*\}$ by final state.

Solution:

$$A = (\{q_0, q_1, q_f\}, \{a, b, c\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \{q_f\})$$

is a pda, where δ is defined as

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}, \quad \delta(q_0, b, Z_0) = \{(q_0, bZ_0)\} \quad (7.5)$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}, \quad \delta(q_0, b, a) = \{(q_0, ba)\} \quad (7.6)$$

$$\delta(q_0, a, b) = \{(q_0, ab)\}, \quad \delta(q_0, b, b) = \{(q_0, bb)\} \quad (7.7)$$

$$\begin{aligned} \delta(q_0, c, a) &= \{(q_1, a)\}, & \delta(q_0, c, b) &= \{(q_1, b)\}, & \delta(q_0, c, Z_0) \\ & & & & = \{(q_1, Z_0)\} \end{aligned} \quad (7.8)$$

$$\delta(q_1, a, a) = \delta(q_1, b, b) = \{(q_1, \Lambda)\} \quad (7.9)$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_f, Z_0)\} \quad (7.10)$$

Acceptance by PDA

Let $wcw^T \in L$. Write $w = a_1a_2 \dots a_n$,
where each a_i is either a or b . Then, we have

$$\begin{aligned}
 & (q_0, a_1a_2 \dots a_n cw^T, Z_0) \\
 & \xrightarrow{*} (q_0, cw^T, a_na_{n-1} \dots a_1Z_0) && \text{by Rules (7.5)–(7.7)} \\
 & \xrightarrow{*} (q_1, a_na_{n-1} \dots a_1, a_na_{n-1} \dots a_1Z_0) && \text{by Rule (7.8)} \\
 & \xrightarrow{\circ} (q_1, \Lambda, Z_0) && \text{by Rule (7.9)} \\
 & \vdash (q_f, \Lambda, Z_0) && \text{by Rule (7.10)}
 \end{aligned}$$

Therefore, $wcw^T \in T(A)$, i.e. $L \subseteq T(A)$.

Hence the PDA A accepts L by final state.

Acceptance by PDA

Acceptance by PDA by null store:

Definition 7.7 Let $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a pda. The set $N(A)$ accepted by null store (or empty store) is defined by

$$N(A) = \{w \in \Sigma^* | (q_0, w, Z_0) \xrightarrow{*} (q, \Lambda, \Lambda) \text{ for some } q \in Q\}$$

Note:

In other words, w is in $N(A)$ if A is in initial ID (q_0, w, Z_0) and empties the PDS after processing all the symbols of w . So in defining $N(A)$, we consider the change brought about on PDS by application of w , and not the transition of states.

Acceptance by PDA

Example:

Let us consider the same PDA given in previous example with an additional rule

$$\delta(q_f, \Lambda, Z_0) = \{(q_f, \Lambda)\} \quad (7.11)$$

Then,

$$N(A) = \{wcw^T \mid w \in \{a, b\}^*\}$$

Solution:

From the construction of A , we see that the Rules (7.5)–(7.10) cannot erase Z_0 . We make a provision for erasing Z_0 from PDS by Rule (7.11). By Example 7.2, $wcw^T \in T(A)$ if and only if the pda reaches the ID (q_f, Λ, Z_0) . By (7.11), PDS can be emptied by Λ -moves if and only if the pda reaches the ID (q_f, Λ, Z_0) . Hence,

$$N(A) = \{wcw^T \mid w \in \{a, b\}^*\}$$

Acceptance by PDA

Example:

Consider a PDA A accepting the set of all strings over $\{a, b\}$ with equal number of a 's and b 's.

Solution:

Let

$$A = (\{q\}, [a, b], [Z_0, a, b], \delta, q, Z_0, \emptyset)$$

where δ is defined by the following rules:

$$\delta(q, a, Z_0) = \{(q, aZ_0)\} \quad \delta(q, b, Z_0) = \{(q, bZ_0)\}$$

$$\delta(q, a, a) = \{(q, aa)\} \quad \delta(q, b, b) = \{(q, bb)\}$$

$$\delta(q, a, b) = \{(q, \Lambda)\} \quad \delta(q, b, a) = \{(q, \Lambda)\}$$

$$\delta(q, \Lambda, Z_0) = \{(q, \Lambda)\}$$

Acceptance by PDA

The construction of δ is done in order to match the number of occurrences of a and b.

We start by storing a symbol of the input string and continue storing until the other symbol occurs.

If the topmost symbol in PDS is a and the current input symbol is b, a in PDS is erased.

If w has equal number of a's and b's

$$(q, w, Z_0) \xrightarrow{*} (q, \Lambda, Z_0) \vdash (q, \Lambda, \Lambda).$$

So, $w \in N(A)$. We can show that $N(A)$ is the given set of strings over $\{a,b\}$ using the construction of δ .

Acceptance by PDA

For example, let the string $w = aabb$

$$\begin{aligned}(q, w, Z_0) \quad &|-- (q, aabb, Z_0) \\ &|-- (q, abb, aZ_0) \\ &|-- (q, bb, aaZ_0) \\ &|-- (q, b, aZ_0) \\ &|-- (q, \Lambda, Z_0) \\ &|-- (q, \Lambda, \Lambda)\end{aligned}$$

Here $w \in N(A)$

Acceptance by PDA

For example, let the string $w = aabbb$

$$(q, w, Z_0) \quad | \dashv (q, aabbb, Z_0)$$
$$| \dashv (q, abbb, aZ_0)$$
$$| \dashv (q, bbb, aaZ_0)$$
$$| \dashv (q, bb, aZ_0)$$
$$| \dashv (q, b, Z_0)$$
$$| \dashv (q, \Lambda, bZ_0)$$

Here the PDA halts.

Push Down Automata and Context-Free Languages

- **CFG to PDA:**

Let $L = L(G)$, where $G = (V_N, \Sigma, P, S)$ is a CFG.

We construct a PDA A as

$$A = (\{q\}, \Sigma, V_N \cup \Sigma, \delta, q, S, \phi)$$

where δ is defined by the following rules:

$$R_1 : \delta(q, \Lambda, A) = \{(q, \alpha) \mid A \rightarrow \alpha \text{ is in } P\}$$

$$R_2 : \delta(q, a, a) = \{(q, \Lambda)\} \text{ for every } a \text{ in } \Sigma$$

Push Down Automata and Context-Free Languages

Example 1:

Construct a PDA A equivalent to the following CFG

$$S \rightarrow 0BB$$

$$B \rightarrow 0S \mid 1S \mid 0$$

Also test whether 010^4 is in $N(A)$.

Push Down Automata and Context-Free Languages

Solution:

The given CFG is in GNF.

Let us define PDA A as follows:

$$A = (\{q\}, \{0, 1\}, \{S, B, 0, 1\}, \delta, q, S, \phi)$$

where δ is defined as

By R_1

$$\delta(q, \Lambda, S) = \{(q, 0BB)\} \quad \text{-----(1)}$$

$$\delta(q, \Lambda, B) = \{(q, 0S), (q, 1S), (q, 0)\} \quad \text{-----(2)}$$

By R_2

$$\delta(q, 0, 0) = \{(q, \Lambda)\} \quad \text{-----(3)}$$

$$\delta(q, 1, 1) = \{(q, \Lambda)\} \quad \text{-----(4)}$$

Push Down Automata and Context-Free Languages

The given string is 010^4

$(q, 010^4, S)$	$ \rightarrow (q, 010^4, 0BB)$	by (1)
	$ \rightarrow (q, 10^4, BB)$	by (3)
	$ \rightarrow (q, 10^4, 1SB)$	by (2)
	$ \rightarrow (q, 0^4, SB)$	by (4)
	$ \rightarrow (q, 0^4, 0BBB)$	by (1)
	$ \rightarrow (q, 0^3, BBB)$	by (3)
	$ \rightarrow (q, 0^3, 0BB)$	by (2)
	$ \rightarrow (q, 0^2, BB)$	by (3)
	$ \rightarrow (q, 0^2, 0B)$	by (2)
	$ \rightarrow (q, 0, B)$	by (3)
	$ \rightarrow (q, 0, 0)$	by (2)
	$ \rightarrow (q, \Lambda, \Lambda)$	by (3)

Thus $010^4 \in N(A)$

Push Down Automata and Context-Free Languages

Example 2:

Construct a PDA A equivalent to the following CFG

$$S \rightarrow aSb \mid A$$

$$A \rightarrow bSa \mid S \mid \Lambda$$

Also test whether a^2b^2 is in $N(A)$.

Push Down Automata and Context-Free Languages

Solution:

The given CFG is in GNF.

Let us define PDA A as follows:

$$A = (\{q\}, \{a, b\}, \{S, A, a, b\}, \delta, q, S, \phi)$$

where δ is defined as

By R_1

$$\delta(q, \Lambda, S) = \{(q, aSb), (q, A)\} \quad \text{-----(1)}$$

$$\delta(q, \Lambda, A) = \{(q, bSa), (q, S), (q, \Lambda)\} \quad \text{-----(2)}$$

By R_2

$$\delta(q, a, a) = \{(q, \Lambda)\} \quad \text{-----(3)}$$

$$\delta(q, b, b) = \{(q, \Lambda)\} \quad \text{-----(4)}$$

Push Down Automata and Context-Free Languages

The given string is a^2b^2

(q, a^2b^2, S)	$ \rightarrow (q, a^2b^2, aSb)$	by (1)
	$ \rightarrow (q, ab^2, Sb)$	by (3)
	$ \rightarrow (q, ab^2, aSbb)$	by (1)
	$ \rightarrow (q, b^2, Sbb)$	by (3)
	$ \rightarrow (q, b^2, Abb)$	by (1)
	$ \rightarrow (q, b^2, bb)$	by (2)
	$ \rightarrow (q, b, b)$	by (4)
	$ \rightarrow (q, \Lambda, \Lambda)$	by (4)

Thus $a^2b^2 \in N(A)$

Push Down Automata and Context-Free Languages

Example 3:

Construct a PDA A equivalent to the following CFG

$$S \rightarrow aA$$

$$A \rightarrow aABC \mid bB \mid a$$

$$B \rightarrow b$$

$$C \rightarrow c$$

Push Down Automata and Context-Free Languages

Solution:

The given CFG is in GNF.

Let us define PDA A as follows:

$$A = (\{q\}, \{a, b, c\}, \{S, A, B, C, a, b, c\}, \delta, q, S, \phi)$$

where δ is defined as

$$\begin{aligned}\text{By } R_1 : \quad & \delta(q, \Lambda, S) = \{(q, aA)\} \\ & \delta(q, \Lambda, A) = \{(q, aABC), (q, bB), (q, a)\} \\ & \delta(q, \Lambda, B) = \{(q, b)\} \\ & \delta(q, \Lambda, C) = \{(q, c)\}\end{aligned}$$

$$\begin{aligned}\text{By } R_2 : \quad & \delta(q, a, a) = \{(q, \Lambda)\} \\ & \delta(q, b, b) = \{(q, \Lambda)\} \\ & \delta(q, c, c) = \{(q, \Lambda)\}\end{aligned}$$

Push Down Automata and Context-Free Languages

Example 4:

Construct a PDA A equivalent to the following CFG

$$S \rightarrow aSbb \mid a$$

Push Down Automata and Context-Free Languages

Solution:

The given CFG is not in GNF. So we first transform into GNF.
The GNF is

$$S \rightarrow aSA \mid a$$

$$A \rightarrow bB$$

$$B \rightarrow b$$

Now, let us define PDA A as follows:

$$A = (\{q\}, \{a, b\}, \{S, A, B, a, b\}, \delta, q, S, \phi)$$

where δ is defined as

$$\text{By } R_1 : \quad \delta(q, \Lambda, S) = \{(q, aSA), (q, a)\}$$

$$\delta(q, \Lambda, A) = \{(q, bB)\}$$

$$\delta(q, \Lambda, B) = \{(q, b)\}$$

$$\text{By } R_2 : \quad \delta(q, a, a) = \{(q, \Lambda)\}$$

$$\delta(q, b, b) = \{(q, \Lambda)\}$$

Push Down Automata and Context-Free Languages

- **PDA to CFG:**

Let $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a PDA.

The construction of CFG G is given as follows:

We define $G = (V_N, \Sigma, P, S)$ where

$$V_N = \{S\} \cup \{[q, Z, q'] \mid q, q' \in Q, Z \in \Gamma\}$$

i.e., any element of V_N is either the new symbol S acting as the start symbol for G or an ordered triple whose first and third elements are states and the second element is a pushdown symbol.

Push Down Automata and Context-Free Languages

The productions in P are induced by moves of PDA as follows:

R_1 : S-productions are given by $S \rightarrow [q_0, Z_0, q]$ for every q in Q .

R_2 : Each move erasing a pushdown symbol given by $(q', \Lambda) \in \delta(q, a, Z)$ induces the production $[q, Z, q'] \rightarrow a$.

Push Down Automata and Context-Free Languages

R_3 : Each move not erasing a push down symbol given by $(q_1, Z_1 Z_2 \dots Z_m) \in \delta(q, a, Z)$ induce many productions of the form

$$[q, Z, q'] \rightarrow a[q_1, Z_1, q_2][q_2, Z_2, q_3] \dots [q_m, Z_m, q']$$

where each of the states $q', q_2, \dots, q_m$ can be any state in Q .

Each move yields many productions because of R_3 .

Push Down Automata and Context-Free Languages

Example 1:

Construct a CFG G which accepts $N(A)$ where

$$A = (\{q_0, q_1\}, \{a, b\}, \{Z_0, Z\}, \delta, q_0, Z_0, \phi)$$

and δ is given by

$$\delta(q_0, b, Z_0) = \{(q_0, ZZ_0)\}$$

$$\delta(q_0, \Lambda, Z_0) = \{(q_0, \Lambda)\}$$

$$\delta(q_0, b, Z) = \{(q_0, ZZ)\}$$

$$\delta(q_0, a, Z) = \{(q_1, Z)\}$$

$$\delta(q_1, b, Z) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, a, Z_0) = \{(q_0, Z_0)\}$$

Push Down Automata and Context-Free Languages

Solution:

Let $G = \{V_N, \{a, b\}, P, S\}$

where V_N consists of

$S, [q_0, Z_0, q_0], [q_0, Z_0, q_1], [q_0, Z, q_0], [q_0, Z, q_1],$
 $[q_1, Z_0, q_0], [q_1, Z_0, q_1], [q_1, Z, q_0], [q_1, Z, q_1]$

The productions are

$$P_1 : S \rightarrow [q_0, Z_0, q_0]$$

$$P_2 : S \rightarrow [q_0, Z_0, q_1]$$

Push Down Automata and Context-Free Languages

$\delta(q_0, b, Z_0) = \{(q_0, ZZ_0)\}$ gives

$$P_3 : [q_0, Z_0, q_0] \rightarrow b[q_0, Z, q_0][q_0, Z_0, q_0]$$

$$P_4 : [q_0, Z_0, q_0] \rightarrow b[q_0, Z, q_1][q_1, Z_0, q_0]$$

$$P_5 : [q_0, Z_0, q_1] \rightarrow b[q_0, Z, q_0][q_0, Z_0, q_1]$$

$$P_6 : [q_0, Z_0, q_1] \rightarrow b[q_0, Z, q_1][q_1, Z_0, q_1]$$

$\delta(q_0, \Lambda, Z_0) = \{(q_0, \Lambda)\}$ gives

$$P_7 : [q_0, Z_0, q_0] \rightarrow \Lambda$$

Push Down Automata and Context-Free Languages

$\delta(q_0, b, Z) = \{(q_0, ZZ)\}$ gives

$$P_8 : [q_0, Z, q_0] \rightarrow b[q_0, Z, q_0][q_0, Z, q_0]$$

$$P_9 : [q_0, Z, q_0] \rightarrow b[q_0, Z, q_1][q_1, Z, q_0]$$

$$P_{10} : [q_0, Z, q_1] \rightarrow b[q_0, Z, q_0][q_0, Z, q_1]$$

$$P_{11} : [q_0, Z, q_1] \rightarrow b[q_0, Z, q_1][q_1, Z, q_1]$$

$\delta(q_0, a, Z) = \{(q_1, Z)\}$ gives

$$P_{12} : [q_0, Z, q_0] \rightarrow a[q_1, Z, q_0]$$

$$P_{13} : [q_0, Z, q_1] \rightarrow a[q_1, Z, q_1]$$

Push Down Automata and Context-Free Languages

$\delta(q_1, b, Z) = \{(q_1, \Lambda)\}$ gives

$$P_{14} : [q_1, Z, q_1] \rightarrow b$$

$\delta(q_1, a, Z_0) = \{(q_0, Z_0)\}$ gives

$$P_{15} : [q_1, Z_0, q_0] \rightarrow a[q_0, Z_0, q_0]$$

$$P_{16} : [q_1, Z_0, q_1] \rightarrow a[q_0, Z_0, q_1]$$

$P_1 - P_{16}$ give the productions in P .

Push Down Automata and Context-Free Languages

Example 2:

Construct a CFG G which accepts $N(A)$ where

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

and δ is given by

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Push Down Automata and Context-Free Languages

Solution:

Let $G = \{V_N, \{a, b\}, P, S\}$

where V_N consists of

$S, [q_0, Z_0, q_0], [q_0, Z_0, q_1], [q_0, a, q_0], [q_0, a, q_1],$
 $[q_1, Z_0, q_0], [q_1, Z_0, q_1], [q_1, a, q_0], [q_1, a, q_1]$

The productions are

$$P_1 : S \rightarrow [q_0, Z_0, q_0]$$

$$P_2 : S \rightarrow [q_0, Z_0, q_1]$$

Push Down Automata and Context-Free Languages

$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$ gives

$$P_3 : [q_0, Z_0, q_0] \rightarrow a[q_0, a, q_0][q_0, Z_0, q_0]$$

$$P_4 : [q_0, Z_0, q_0] \rightarrow a[q_0, a, q_1][q_1, Z_0, q_0]$$

$$P_5 : [q_0, Z_0, q_1] \rightarrow a[q_0, a, q_0][q_0, Z_0, q_1]$$

$$P_6 : [q_0, Z_0, q_1] \rightarrow a[q_0, a, q_1][q_1, Z_0, q_1]$$

$\delta(q_0, a, a) = \{(q_0, aa)\}$ gives

$$P_7 : [q_0, a, q_0] \rightarrow a[q_0, a, q_0][q_0, a, q_0]$$

$$P_8 : [q_0, a, q_0] \rightarrow a[q_0, a, q_1][q_1, a, q_0]$$

$$P_9 : [q_0, a, q_1] \rightarrow a[q_0, a, q_0][q_0, a, q_1]$$

$$P_{10} : [q_0, a, q_1] \rightarrow a[q_0, a, q_1][q_1, a, q_1]$$

Push Down Automata and Context-Free Languages

$\delta(q_0, b, a) = \{(q_1, a)\}$ gives

$$P_{11} : [q_0, a, q_0] \rightarrow b[q_1, a, q_0]$$

$$P_{12} : [q_0, a, q_1] \rightarrow b[q_1, a, q_1]$$

$\delta(q_1, b, a) = \{(q_1, a)\}$ gives

$$P_{13} : [q_1, a, q_0] \rightarrow b[q_1, a, q_0]$$

$$P_{14} : [q_1, a, q_1] \rightarrow b[q_1, a, q_1]$$

$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$ gives

$$P_{15} : [q_1, a, q_1] \rightarrow a$$

$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$ gives

$$P_{16} : [q_1, Z_0, q_1] \rightarrow \Lambda$$

$P_1 - P_{16}$ give the productions in P .

Deterministic Push Down Automata (DPDA)

Example 1:

Design a PDA for

$$L = \{ w \in \{a, b\}^* \mid w \text{ is of the form } a^n b^n \text{ s.t. } n \geq 1 \}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = a^3b^3$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

- Sequence of a's should be pushed onto the stack.
- For every b's, a's should be popped from the stack.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, Z_0)\} \text{ or } \{(q_2, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \{q_2\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = a^3b^3$

$(q_0, aaabbb, Z_0)$	--	$(q_0, aabbb, aZ_0)$
	--	$(q_0, abbb, aaZ_0)$
	--	$(q_0, bbb, aaaZ_0)$
	--	(q_1, bb, aaZ_0)
	--	(q_1, b, aZ_0)
	--	(q_1, Λ, Z_0)
	--	(q_2, Λ, Λ)

Therefore, a^3b^3 is accepted by final state.

Deterministic Push Down Automata (DPDA)

Example 2:

Design a PDA for

$$L = \{a^m b^n \mid n < m\}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = a^3 b^2$ is accepted by empty store or not.

Deterministic Push Down Automata (DPDA)

Solution:

- Sequence of a's should be pushed onto the stack.
- For every b's, a's should be popped from the stack.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, a) = \{(q_1, \Lambda)\}$$

[For additional m-n a's should be popped from stack]

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, a) = \{(q_1, \Lambda)\}$$

[For additional m-n a's should be popped from stack]

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, Z_0)\} \text{ or } \{(q_2, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \{q_2\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = a^3b^2$

$(q_0, aaabb, Z_0)$	--	$(q_0, aabb, aZ_0)$
	--	(q_0, abb, aaZ_0)
	--	$(q_0, bb, aaaZ_0)$
	--	(q_1, b, aaZ_0)
	--	(q_1, Λ, aZ_0)
	--	(q_1, Λ, Z_0)
	--	(q_1, Λ, Λ)

Therefore, a^3b^2 is accepted by empty store.

Deterministic Push Down Automata (DPDA)

Example 3:

Design a PDA for accepting the set of all strings over $\{a, b\}$ with an equal number of a's and b's for both empty store and final state.

Also check whether $w = abbaab$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb)\}$$

$$\delta(q_0, a, b) = \{(q_0, \Lambda)\}$$

$$\delta(q_0, b, a) = \{(q_0, \Lambda)\}$$

$$\delta(q_0, \Lambda, Z_0) = \{(q_0, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0\}, \{a, b\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb)\}$$

$$\delta(q_0, a, b) = \{(q_0, \Lambda)\}$$

$$\delta(q_0, b, a) = \{(q_0, \Lambda)\}$$

$$\delta(q_0, \Lambda, Z_0) = \{(q_1, \Lambda)\} \text{ or } \{(q_1, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \{q_1\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = abbaab$

$(q_0, abbaab, Z_0)$	--	$(q_0, bbaab, aZ_0)$
	--	$(q_0, baab, Z_0)$
	--	(q_0, aab, bZ_0)
	--	(q_0, ab, Z_0)
	--	(q_0, b, aZ_0)
	--	(q_0, Λ, Z_0)
	--	(q_1, Λ, Λ)

Therefore, $abbaab$ is accepted by final state.

Deterministic Push Down Automata (DPDA)

Example 4:

Design a PDA for accepting $L = \{a^n b^m a^n \mid m, n \geq 1\}$ by empty store and final state.

Also check whether $w = a^2 b a^2$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, \Lambda)\} \text{ or } \{(q_2, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \{q_2\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = a^2ba^2$

$(q_0, aabaa, Z_0)$	--	$(q_0, abaa, aZ_0)$
	--	(q_0, baa, aaZ_0)
	--	(q_1, aa, aaZ_0)
	--	(q_1, a, aZ_0)
	--	(q_1, Λ, Z_0)
	--	(q_2, Λ, Λ)

Therefore, aabaa is accepted by final state.

Deterministic Push Down Automata (DPDA)

Example 5:

Design a PDA for

$$L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = abc^2$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

- For every input symbol 'a', a symbol 'X' is pushed onto the stack.
- For every input symbol 'b', a symbol 'X' is pushed onto the stack.
- For every input symbol 'c', a symbol 'X' is popped from the stack.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, XZ_0)\}$$

$$\delta(q_0, a, X) = \{(q_0, XX)\}$$

$$\delta(q_0, b, Z_0) = \{(q_1, XZ_0)\}$$

$$\delta(q_1, b, X) = \{(q_1, XX)\}$$

$$\delta(q_0, b, X) = \{(q_1, XX)\}$$

$$\delta(q_0, c, X) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, X) = \{(q_1, \Lambda)\}$$

$$\delta(q_0, \Lambda, Z_0) = \{(q_0, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b, c\}, \{X, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, XZ_0)\}$$

$$\delta(q_0, a, X) = \{(q_0, XX)\}$$

$$\delta(q_0, b, Z_0) = \{(q_1, XZ_0)\}$$

$$\delta(q_1, b, X) = \{(q_1, XX)\}$$

$$\delta(q_0, b, X) = \{(q_1, XX)\}$$

$$\delta(q_0, c, X) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, X) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, c, X) = \{(q_2, \Lambda)\}$$

$$\delta(q_0, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, \Lambda, Z_0) = \{(q_3, \Lambda)\} \text{ or } \{(q_3, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2, q_3\}, \{a, b, c\}, \{X, Z_0\}, \delta, q_0, Z_0, \{q_3\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = abc^2$

$(q_0, abcc, Z_0)$	--	(q_0, bcc, XZ_0)
	--	(q_1, cc, XXZ_0)
	--	(q_2, c, XZ_0)
	--	(q_2, Λ, Z_0)
	--	(q_3, Λ, Λ)

Therefore, abc^2 is accepted by final state.

Deterministic Push Down Automata (DPDA)

Example 6:

Design a PDA for

$$L = \{a^n b^n c^m d^m \mid n, m \geq 1\}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = abc^2d^2$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

- Sequence of a's should be pushed onto the stack.
- For every b's, a's should be popped from the stack.
- Sequence of c's should be pushed onto the stack.
- For every d's, c's should be popped from the stack.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, Z_0) = \{(q_1, cZ_0)\}$$

$$\delta(q_1, c, c) = \{(q_1, cc)\}$$

$$\delta(q_1, d, c) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, d, c) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, \Lambda, Z_0) = \{(q_2, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b, c, d\}, \{a, c, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, Z_0) = \{(q_1, cZ_0)\}$$

$$\delta(q_1, c, c) = \{(q_1, cc)\}$$

$$\delta(q_1, d, c) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, d, c) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, \Lambda, Z_0) = \{(q_3, \Lambda)\} \text{ or } \{(q_3, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2, q_3\}, \{a, b, c, d\}, \{a, c, Z_0\}, \delta, q_0, Z_0, \{q_3\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = abc^2d^2$

$(q_0, abccdd, Z_0)$	--	$(q_0, bccdd, aZ_0)$
	--	$(q_1, ccdd, Z_0)$
	--	(q_1, cdd, cZ_0)
	--	(q_1, dd, ccZ_0)
	--	(q_1, cdd, cZ_0)
	--	(q_2, Λ, Z_0)
	--	(q_3, Λ, Λ)

Therefore, abc^2 is accepted by final state.

Deterministic Push Down Automata (DPDA)

Example 7:

Design a PDA for

$$L = \{a^n b^{2n} \mid n \geq 0\}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = a^2b^4$ is accepted by empty store or not.

Deterministic Push Down Automata (DPDA)

Solution:

- Sequence of a's should be pushed onto the stack.
- For every pair of b's, one 'a' should be popped from the stack.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, b, a) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, b, a) = \{(q_1, a)\}$$

$$\delta(q_2, \Lambda, Z_0) = \{(q_2, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_1, b, a) = \{(q_2, \Lambda)\}$$

$$\delta(q_2, b, a) = \{(q_1, a)\}$$

$$\delta(q_2, \Lambda, Z_0) = \{(q_3, \Lambda)\} \text{ or } \{(q_3, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \{q_3\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = a^2b^4$

$(q_0, aabbbb, Z_0)$	--	$(q_0, abbbb, aZ_0)$
	--	$(q_0, bbbb, aaZ_0)$
	--	(q_1, bbb, aaZ_0)
	--	(q_2, bb, aZ_0)
	--	(q_1, b, aZ_0)
	--	(q_2, Λ, Z_0)
	--	(q_2, Λ, Λ)

Therefore, a^2b^4 is accepted by empty store.

Deterministic Push Down Automata (DPDA)

Example 8:

Design a PDA for

$$L = \{a^m b^m c^n \mid n, m \geq 1\}$$

- i. Accepting L by Empty Store
- ii. Accepting L by Final State

Also check whether $w = aabbccc$ is accepted by final state or not.

Deterministic Push Down Automata (DPDA)

Solution:

- Sequence of a's should be pushed onto the stack.
- For every b's, a's should be popped from the stack.
- For every c's, stack is retained as it is.

Deterministic Push Down Automata (DPDA)

i. By Empty Store:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, Z_0) = \{(q_1, Z_0)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b, c\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

ii. By Final State:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, c, Z_0) = \{(q_1, Z_0)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_2, \Lambda)\} \text{ or } \{(q_2, Z_0)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1, q_2\}, \{a, b, c\}, \{a, Z_0\}, \delta, q_0, Z_0, \{q_2\})$$

where δ is given as above.

Deterministic Push Down Automata (DPDA)

Given $w = aabbccc$

$(q_0, aabbccc, Z_0)$	--	$(q_0, abbccc, aZ_0)$
	--	$(q_0, bbccc, aaZ_0)$
	--	$(q_1, bccc, aZ_0)$
	--	(q_1, ccc, Z_0)
	--	(q_1, cc, Z_0)
	--	(q_1, c, Z_0)
	--	(q_1, Λ, Z_0)
	--	(q_2, Λ, Λ)

Therefore, $aabbccc$ is accepted by final state.

Non-Deterministic Push Down Automata (NPDA)

- In a DPDA there is only one move in every situation, whereas, in case of NPDA there could be multiple moves under a situation.
- DPDA is less powerful than NPDA. Every context-free language cannot be recognized by a DPDA but it can be recognized by NPDA.
- The class of language a DPDA can accept lies in between a regular language and CFL (context-free language).
- **Note:**
 - A palindrome can be accepted by NPDA but it cannot be accepted by a DPDA.

Non-Deterministic Push Down Automata (NPDA)

Example 1:

Design a PDA for odd length palindrome over $\{a,b\}$.

Also check whether $w = ababa$ is accepted or not.

Non-Deterministic Push Down Automata (NPDA)

Solution:

- An odd palindrome will be of the form:
 waw^R or wbw^R
- Every character should be considered for two cases:
not a middle character and
middle character
- Since there is no way of finding the middle position of a string by a PDA, therefore, the middle position is fixed non-deterministically.

Non-Deterministic Push Down Automata (NPDA)

Case I: Not middle character:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb)\}$$

$$\delta(q_0, b, a) = \{(q_0, ba)\}$$

$$\delta(q_0, a, b) = \{(q_0, ab)\}$$

Non-Deterministic Push Down Automata (NPDA)

Case II: Middle character:

$$\delta(q_0, a, Z_0) = \{(q_1, Z_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_1, Z_0)\}$$

$$\delta(q_0, a, a) = \{(q_1, a)\}$$

$$\delta(q_0, b, b) = \{(q_1, b)\}$$

$$\delta(q_0, b, a) = \{(q_1, a)\}$$

$$\delta(q_0, a, b) = \{(q_1, b)\}$$

Non-Deterministic Push Down Automata (NPDA)

For popping:

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, b) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Non-Deterministic Push Down Automata (NPDA)

On combining all the cases PDA for odd length palindrome is given as follows:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0), (q_1, Z_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0), (q_1, Z_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa), (q_1, a)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb), (q_1, b)\}$$

$$\delta(q_0, b, a) = \{(q_0, ba), (q_1, a)\}$$

$$\delta(q_0, a, b) = \{(q_0, ab), (q_1, b)\}$$

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, b) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Non-Deterministic Push Down Automata (NPDA)

Given $w = ababa$

$(q_0, ababa, Z_0)$	--	$(q_0, baba, aZ_0)$
	--	(q_0, aba, baZ_0)
	--	(q_1, ba, baZ_0)
	--	(q_1, a, aZ_0)
	--	(q_1, Λ, Z_0)
	--	(q_1, Λ, Λ)

Therefore, $ababa$ is accepted.

Non-Deterministic Push Down Automata (NPDA)

Example 2:

Design a PDA for even length palindrome over $\{a,b\}$.

Also check whether $w = abba$ is accepted or not.

Non-Deterministic Push Down Automata (NPDA)

Solution:

- Here the string is of the form: ww^R
- There is no way of finding the middle position of a string by a PDA, therefore, the middle position is fixed non-deterministically.

Non-Deterministic Push Down Automata (NPDA)

Case I: It belongs to 1st half:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb)\}$$

$$\delta(q_0, b, a) = \{(q_0, ba)\}$$

$$\delta(q_0, a, b) = \{(q_0, ab)\}$$

Case II: It is 1st character of 2nd half:

$$\delta(q_0, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_0, b, b) = \{(q_1, \Lambda)\}$$

Non-Deterministic Push Down Automata (NPDA)

After the state is changed to q_1 , the possible moves are

For popping:

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, b) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Non-Deterministic Push Down Automata (NPDA)

On combining all the cases PDA for even length palindrome is given as follows:

$$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$$

$$\delta(q_0, b, Z_0) = \{(q_0, bZ_0)\}$$

$$\delta(q_0, a, a) = \{(q_0, aa), (q_1, \Lambda)\}$$

$$\delta(q_0, b, b) = \{(q_0, bb), (q_1, \Lambda)\}$$

$$\delta(q_0, b, a) = \{(q_0, ba)\}$$

$$\delta(q_0, a, b) = \{(q_0, ab)\}$$

$$\delta(q_1, a, a) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, b, b) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, \Lambda, Z_0) = \{(q_1, \Lambda)\}$$

Therefore, PDA A is defined as

$$A = (\{q_0, q_1\}, \{a, b\}, \{a, b, Z_0\}, \delta, q_0, Z_0, \phi)$$

where δ is given as above.

Non-Deterministic Push Down Automata (NPDA)

Given $w = abba$

$(q_0, abba, Z_0)$	$ \rightarrow$	(q_0, bba, aZ_0)
	$ \rightarrow$	(q_0, ba, baZ_0)
	$ \rightarrow$	(q_1, a, aZ_0)
	$ \rightarrow$	(q_1, Λ, Z_0)
	$ \rightarrow$	(q_1, Λ, Λ)

Therefore, $abba$ is accepted.

Two – Stack PDA

- A two – stack PDA is similar to a PDA, but it has **two stacks** instead of one.
- A two – stack PDA is deterministic and its move takes into account the symbols on top of both stacks.
- It can push or pop on either or both the stacks.
- Each move of a two – stack PDA will be determined by four things:
 - Current state
 - The next input
 - The symbol on top of first stack
 - The symbol on top of second stack

Two – Stack PDA

Example:

Design a two – stack PDA to accept the language

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

Two – Stack PDA

Solution:

- Sequence of a's is pushed onto both stacks.
- For every b's, a's is popped from 1st stack and a's is retained in 2nd stack.
- For every c's, a's is popped from 2nd stack and topmost symbol of 1st stack is retained.

Two – Stack PDA

The transition function δ is given by

$$\delta(q_0, a, Z_0, Z_0) = \{(q_0, aZ_0, aZ_0)\}$$

$$\delta(q_0, a, a, a) = \{(q_0, aa, aa)\}$$

$$\delta(q_0, b, a, a) = \{(q_1, \Lambda, a)\}$$

$$\delta(q_1, b, a, a) = \{(q_1, \Lambda, a)\}$$

$$\delta(q_1, c, Z_0, a) = \{(q_2, Z_0, \Lambda)\}$$

$$\delta(q_2, c, Z_0, a) = \{(q_2, Z_0, \Lambda)\}$$

$$\delta(q_2, \Lambda, Z_0, Z_0) = \{(q_2, \Lambda, \Lambda)\}$$